

Power Analysis

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1 Power Analysis 1

Design Scenario: Expert vs. Influencer (2 treatment + 1 control) (written stimuli)

We select tau based on small standardized mean difference (d=0.2) as a detectable benchmark as used in Case et al, 2022 with a 7-point Likert outcomes.

- tau_expert = .4
- tau_influencer = .2

The model takes the form:

$$Y_i = \beta_0 + \beta_1 * Expert + \beta_2 * Influencer + \epsilon_i$$

```
print(power_1)
```

##	sample_pct	N	expert_power	influencer_power
##	<num>	<num>	<num>	<num>
## 1:	0.1	50	0.069	0.044
## 2:	0.5	250	0.246	0.102
## 3:	1.0	500	0.153	0.153
## 4:	1.3	650	0.513	0.192
## 5:	2.0	1000	0.752	0.278
## 6:	3.0	1500	0.905	0.333
## 7:	5.0	2500	0.984	0.490

With 10% (50 obs) of recruitment size of the original data we have a power of 6.9%, and 4.4% for expert and influencer group respectively. Given our largest simulation with (2500 obs) we obtain a power of 98.4%, and 49% for expert and influencer group respectively.

2 Power Analysis 2

Design scenario: 3x3 factorial between-subjects design testing the impact of source cues (expert, influencer, control) and stimulus type (video, written, both) on the credibility of nutrition recommendations. It includes a pre-test and post-test that measures the credibility of recommendations before and after the treatment. Power is estimated from OLS on post-test with pre-test as a covariate, using cluster-robust standard errors by participant. Suggested effect sizes in parameters section below (see Section 4 for justification).

```
# Display results
print(power_results)
```

##	sample_size	power_expert	power_influencer
## 1	50	0.126	0.128
## 2	100	0.165	0.156
## 3	150	0.208	0.215
## 4	200	0.253	0.240
## 5	250	0.322	0.323
## 6	300	0.339	0.377
## 7	350	0.413	0.403
## 8	400	0.430	0.444
## 9	450	0.511	0.475
## 10	500	0.539	0.539

3 Power Analysis 3

We are designing a multi-factor design to assess the power using monte carlo simulation Factors to consider are: Source: Expert vs Influencer Content: Message vs Video or Both

Cells:

- Expert-Message
- Expert-Video
- Expert-Both
- Influencer-Message
- Influencer-Video
- Influencer-Both

Now pick the cell means that encode the interaction we care about Example “true” means:

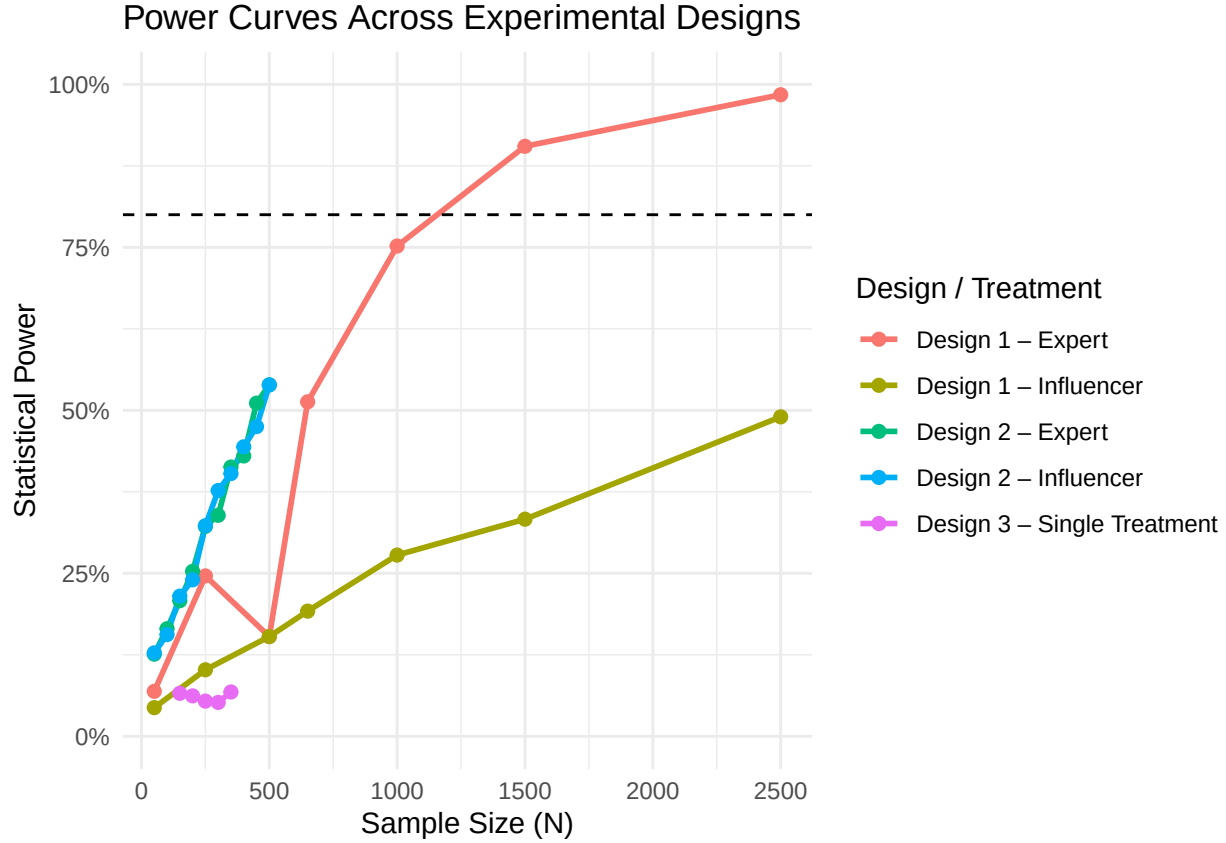
- Expert-Message - 4.8
- Expert-Video - 5.0
- Expert-Both - 5.5 (Expert video and message does slightly better)
- Influencer-Message - 4.2
- Influencer-Video - 4.6
- Influencer-Both - 4.9

```
print(power_grid)
```

```
##      n power conv_rt
## 1 150 0.066      1
## 2 200 0.062      1
## 3 250 0.054      1
## 4 300 0.052      1
## 5 350 0.068      1
```

4 Combined Results

```
combined_plot
```



5 Justification of Effect Sizes

Studies of the impact of source cues (expert vs. non-expert or influencer) on the credibility of information find mixed effects. The studies reviewed examined the impact of source cues in the context of health information (COVID, e-cigarettes), nutrition, politics, and the environment, and generally found that context impacted the effect of treatment. Lee et al. (2024) found a 0.84 increase in the trust in a health message when delivered by an expert (vs. influencer), and Hubner & Watts (2024) found a 0.49 increase in credibility of health information when delivered by an expert (vs. celebrity). In contrast, Case et al. (2022) found no significant effect of treatment because covariates (specifically political ideology) explained much of the variation in their data. In addition, Lissens et al. (2025) also found no effect of source cues on nutrition messaging to young people. Given this range, we suggest the effect size of our proposed treatment may be around 0.3 (slightly higher in a design that decreases standard errors). We believe that it will be lower than that found in health messaging, but potentially higher than Lissens found with a youth audience as nutritional messaging may be more salient for a mixed-age audience who might be more worried about the consequences of their choices.

6 Bibliography

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